

MBA 753: Causal Inference Models

Assignment 1

Note: Please name the file as follows: “your roll number - your first name”, for example: “123456789 - Nivedita”. Save it as a pdf and submit.

Q1. Using UNHDD data, fit a regression model for GNI (Gross National Income) with the partial effects of HDI (Human Development Index), GII (Gender Inequality Index), life expectancy at birth, and population age 15-64 years (in millions). **[20 marks]**

- a) Which effects are significant? **[2 marks]**
- b) Perform model diagnostics. Mention the assumptions tested, methods used and their corresponding results. **[10 marks]**
- c) Interpret the regression coefficients. **[8 marks]**

Q2. In the above regression model, replace HDI with HDI groups (all other predictors have to be retained in the model). Use “low” as the reference category for HDI groups. **[20 marks]**

- a) How many categories of HDI group are present in the dataset? **[1 mark]**
- b) Interpret the regression coefficients of HDI groups? Are they in tune with the results from question 1? **[9 marks]**
- c) Change the reference group to “Very High” and fit the model. Interpret the regression coefficients of the categorical predictor. What are your conclusions about the model? **[10 marks]**

Q3. In the regression model from question 2, include an interaction term between HDI groups and GII. **[20 marks]**

- a) Is the interaction significant? **[3 marks]**
- b) Interpret the interaction effects. **[5 marks]**
- c) Perform model diagnostics and report their results. **[10 marks]**
- d) What are your conclusions about the regression model? **[2 marks]**